INCH-POUND

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SUPERSEDING
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18 May 1998

COMMERCIAL ITEM DESCRIPTION

LEADS, STORAGE-BATTERY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers low tension cables used for connections between storage batteries and other electrical equipment. The insulated leads are used for connections between the battery and equipment where heavy amperage 6-, 12-, and 24-volt (V) ungrounded circuits interconnect. The uninsulated leads and straps are used for bonding and grounding connections.

2. CLASSIFICATION. The leads shall be of the following types, classes, and styles as specified (see 7.3(b)).

Type I - Ground lead, uninsulated strap, flat woven or braided

Style 1 - Battery to ground
Style 2 - Ground to ground (bonding strap)

Type II - Battery lead, insulated (rope strand, concentric lay, or bunched)

Class A - Extra-flexible rope strand
Class B - Stranded concentric lay or bunched

Style 1 - Rope lay, concentric lay, or bunched with generator lead
Style 2 - Rope lay, concentric lay, or bunched with lugs
Style 3 - Rope lay, concentric lay, or bunched with straight universal post terminals
Style 4 - Rope lay, concentric lay, or bunched with angle universal post terminals
Style 5 - Rope lay, concentric lay, or bunched with side terminal and lug
Style 6 - Rope lay, concentric lay, or bunched with side terminals

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

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3. SALIENT CHARACTERISTICS

3.1 Standard commercial product. The storage battery leads shall, as a minimum, be in accordance with the requirements of this CID and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this CID but which are a part of the manufacturer's standard commercial product shall be included in the storage battery leads being furnished.

3.2 Materials. The storage battery leads shall consist of a specified length (see 7.3(c)) of low tension cable defined as a center to center distance between terminal holes (cable length) which is terminated at each end with a post terminal lug as specified.

3.3 Design. Insulator boots or caps shall be provided for type I, style 1 leads and provide an additional margin of safety during installation and servicing of the battery cable connections. Type II, class A or B leads, styles 1 through 6 shall be provided with insulator boots or caps to prevent inadvertent contact with energized conductors. Material used for construction of the cable or lead insulation shall be consistent with standard commercial practice.

3.3.1 Type I. Type I ground leads are uninsulated flat strips and shall be composed of braided copper conductors and plated to resist corrosion.

3.3.1.1 Type I, style 1. Type I, style 1 ground leads shall be plated to resist corrosion and shall have a post terminal (see 3.3.5.1) on one end and a terminal lug (see 3.3.5.3) on the other end. The style 1 lead is used for connection from the battery to ground.

3.3.1.2 Type I, style 2. Type I, style 2 ground leads shall be uncoated and have a terminal lug (see 3.3.5.3) on each end for use from engine to ground.

3.3.2 Type II. Type II insulated leads shall be general purpose, insulated, braidless, and classified as extra flexible rope strand construction (class A), bunched, or stranded concentric lay (class B).

3.3.2.1 Type II, class A. Type II, class A insulated extra flexible rope strand leads are used for connection between the battery and the starter, the battery and the solenoid, the starter and the solenoid, or the battery and the ground.

3.3.2.2 Type II, class B. Type II, class B insulated strand concentric lay or bunched leads are used for connection between the battery and the starter, the battery and the solenoid, the starter and the solenoid, or the battery and the ground.

3.3.3 Cable style. Type II, class A or B cable designs shall be designated by the cable styles.

3.3.3.1 Type II, class A or B, style 1. Type II, class A or B, style 1 leads shall be an insulated generator lead attached to a universal post terminal (see 3.3.5.1). The other end of the lead shall be fitted with a terminal lug (see 3.3.5.3).

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3.3.3.2 Type II, class A or B, style 2. Type II, class A or B, style 2 leads shall have a terminal lug (see 3.3.5.3) on each end. This style lead is used for connection between the solenoid and starter and similar applications.

3.3.3.3 Type II, class A or B, style 3. Type II, class A or B, style 3 leads shall have straight-neck post terminals (see 3.3.5.1) on each end. The post terminals used shall be the universal type fitting either a negative or positive battery post. This style lead is used as a connector when arranging batteries in series or parallel combinations.

3.3.3.4 Type II, class A or B, style 4. Type II, class A or B, style 4 leads shall have angle type post terminals (see 3.3.5.1). The post terminals used on these leads shall be the universal type arranged 180 degrees with respect to each other. This style lead is used as a connector when arranging batteries in series or parallel combinations.

3.3.3.5 Type II, class A or B, style 5. Type II, class A or B, style 5 leads shall have a side terminal with an integral bolt (see 3.3.5.2) on one end and a lug (see 3.3.5.3) on the other end. This style lead is used for connection between a side terminal battery and the starter or between the battery and ground.

3.3.3.6 Type II, class A or B, style 6. Type II, class A or B, style 6 leads shall have a side terminal with an integral bolt (see 3.3.5.2) on each end. This style lead is used as a connector when arranging side terminal batteries in series or parallel combinations.

3.3.4 Insulation protective covering. When additional protective covering, such as waterproof loom, is required for type II lead, the length of the loom and its location on the lead shall be as specified (see 7.3(d)).

3.3.5 Termination. The termination for the leads shall conform to the requirements of 3.3.5.1 through 3.3.5.3, as applicable.

3.3.5.1 Lead terminals (battery post). The lead terminals for battery posts shall be straight-neck type cast or molded from corrosion-resistant metal. The terminal shall be designed to permit complete tightening with full contact on the battery post. A bolt and nut having corrosion-resistant finish shall be provided with each terminal. The clamping bolt shall permit tightening to the degree necessary to provide a full power connection. The terminal shall be capable of providing a full power connection to either a positive or negative post.

3.3.5.2 Lead terminals (side). Leads having side terminals shall be of correct design to connect to the electrical connection of a side terminal type battery, and shall have a corrosion-resistant finish. Each lead terminal shall include a bolt. The terminal shall be capable of providing a full power connection to either a positive or negative contact of a side terminal type battery.

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3.3.5.3 Terminal lugs.

3.3.5.3.1 Type I. The terminal lugs for type I shall be compressed to fit closely around the flat ground strap and shall be cadmium plated or coated with lead-antimony alloy.

3.3.5.3.2 Type II. The terminal lugs shall for type II be formed from copper or brass alloy and be plated or coated with lead-antimony alloy.

3.3.6 Mechanical strength. All completed cable leads shall be capable of withstanding a pull of 75 pounds (34 kilograms (kg)) for 1 minute.

3.4 Identification data. The lead assembly shall show inscribed on or attached to, the manufacturer's name or trademark and part number. When specified (see 7.3(e)), the lead assembly shall also identify the manufacturer's production lot number.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in ASTM SI-10 (IEEE/ASTM SI-10), "American National Standard for Use of the International System of Units (SI): The Modern Metric System", and all other requirements of this CID including form, fit, and function are met. If a product is manufactured to metric dimensions and these dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

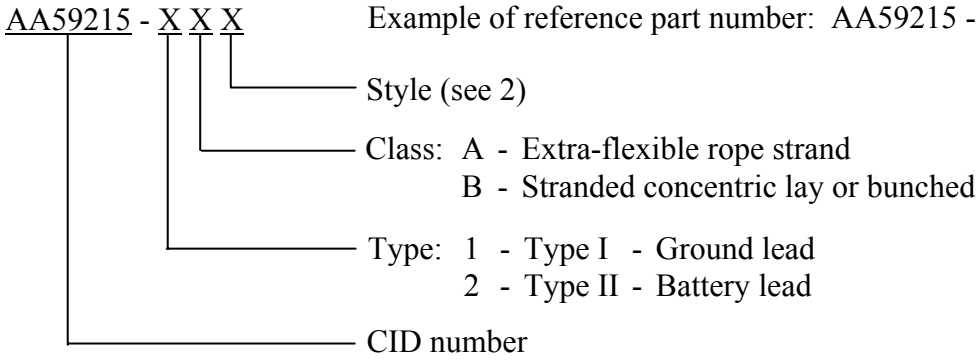
6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(f)).

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7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Lead type, class, and style (see 2).
c. Cable length (see 3.2).
d. Length of the loom and location on the lead when insulation protective covering is required (see 3.3.4).
e. Production lot number, when required (see 3.4).
f. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

concentric lay
insulated
post terminal
rope lay
terminal lug style
uninsulated

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MILITARY INTERESTS:

Custodians:
Army - CR
Navy - YD
Air Force - 99

Review Activities:
Navy - AS, CG
Air Force - 03, 19, 33, 71

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS2

(Project 6150-2005-002)

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